# WebPro Italy

**Web Design Agency - Project Report**

**Web Programming Exam**

**Student: Yasin Bargahi**

**GitHub: meganis10**

**April 2026**

https://github.com/meganis10/webagency-project

## 1. Introduction

For my web programming exam project I decided to build a website for a web design agency called WebPro Italy. I chose this topic because I find web design interesting and I wanted to build something that looks like a real business website.

The website has two sides - a public side where visitors can see the services, register an account and send messages, and an admin side where I can manage everything. I built it using PHP, MySQL, HTML, CSS and JavaScript with jQuery.

I started from scratch and tried to follow good practices I learned, especially for security since the professor mentioned it is important.

## 2. Technologies I Used

| Technology | Why I used it |
|---|---|
| PHP | For the backend - connecting to database, handling forms, building APIs |

| | |
|---|---|
| MySQL | To store users, services, messages and admin accounts |
| HTML5 | To build the structure of all pages |
| CSS3 | To style the pages and make them look good |
| JavaScript + jQuery | For animations, active nav links, fade effects |
| PDO | PHP tool for database - more secure than basic MySQL functions |
| XAMPP | Local server on my computer for development and testing |

I chose PHP because it works well with MySQL and is easy to use for web projects. I used PDO instead of basic MySQL functions because PDO has prepared statements which stop SQL injection. jQuery was added because the professor asked for it and it made animations much easier to write.

## 3. Project Structure

I organized the project into folders to keep things clean and easy to find:

| File / Folder | What it does |
|---|---|
| index.php | Homepage - shows services from database |
| register.php | Registration form with duplicate email check |
| login.php | Login form with password verification |
| logout.php | Destroys session and sends user to homepage |
| profile.php | Shows user info, option to delete account |
| about.php | About the agency page |
| services.php | Shows all services publicly |
| contact.php | Contact form that saves message to database |
| admin/dashboard.php | Admin homepage with statistics |
| admin/services.php | Admin can add or delete services |

| | |
|---|---|
| admin/messages.php | Admin can read and delete messages |
| admin/users.php | Admin can see and delete users |
| admin/admins.php | Admin can manage other admin accounts |
| api/services.php | Returns services as JSON |
| api/register.php | API to register a user |
| api/login.php | API to login a user |
| api/users.php | API to get users list (needs admin key) |
| components/connect.php | Database connection used by all pages |
| components/header.php | Navigation header included in every page |
| components/admin_header.php | Admin header with access control |
| css/style.css | Styles for public pages |
| css/admin.css | Styles for admin panel |
| js/script.js | jQuery code for animations |

I put connect.php, header.php and footer.php in a components folder because every page needs them. This way if I need to change the header I only change one file instead of changing every page.

## 4. Database

The database is called webagency_db and has 4 tables:

| Table | Columns |
|---|---|
| users | id, name, email, password, dob, country, city, created_at |
| admins | id, name, email, password |
| messages | id, name, email, subject, message, created_at |
| services | id, title, description, icon |

For the ID of each record I used random_bytes(16) converted to hex instead of normal numbers like 1, 2, 3. I did this because if I use numbers someone could guess the ID and try to delete other people's records. Random hex IDs are impossible to guess.

The email column in both users and admins tables has UNIQUE constraint so the database itself also prevents duplicate emails, not just the PHP code.

## 5. Security

Security was something I tried to focus on. Here is what I implemented:

**SQL Injection Prevention:** I used PDO prepared statements everywhere. The ? placeholder means user input is never directly put inside the SQL query so it cannot break or change the query.

**Password Hashing:** Passwords are never saved as plain text. I use password_hash() with BCRYPT to convert them before saving. To check a password I use password_verify() which compares without ever reversing the hash.

**Input Sanitization:** All form inputs go through filter_var() with FILTER_SANITIZE_STRING or FILTER_SANITIZE_EMAIL to remove dangerous characters before processing.

**Session Validation:** Every admin page checks two things - first that the session exists, then that the admin account still exists in the database. If an admin is deleted while logged in, they get kicked out on the next page load.

**Access Control:** Admin pages redirect to login if no admin session exists. An admin cannot delete their own account. Users cannot access the admin panel at all.

**API Key:** The users API requires an admin_key in the URL. Without it the response is Unauthorized. This prevents anyone from reading the user list.

## 6. API Endpoints

I created 4 API endpoints in the api/ folder. They all return JSON. I added these because the professor asked for backend APIs and also because it makes the project more complete - a mobile app could use these same APIs.

| URL | Method | What it returns |
|---|---|---|
| api/services.php | GET | List of all services as JSON |
| api/register.php | POST | Registers a user, returns success or error |
| api/login.php | POST | Checks credentials, returns user info or error |

| | | |
|---|---|---|
| api/users.php?admin_key=webpro2025 | GET | List of all users (admin only) |

## 7. What I Learned

This project taught me a lot of things I did not know before. The most important thing I learned is how a real website works behind the scenes - how forms send data to PHP, how PHP talks to MySQL, and how the result comes back as HTML.

I also learned why security is important. Before this project I did not think much about SQL injection or password hashing but now I understand why these things matter. If I saved passwords as plain text and the database got stolen, all users would be at risk.

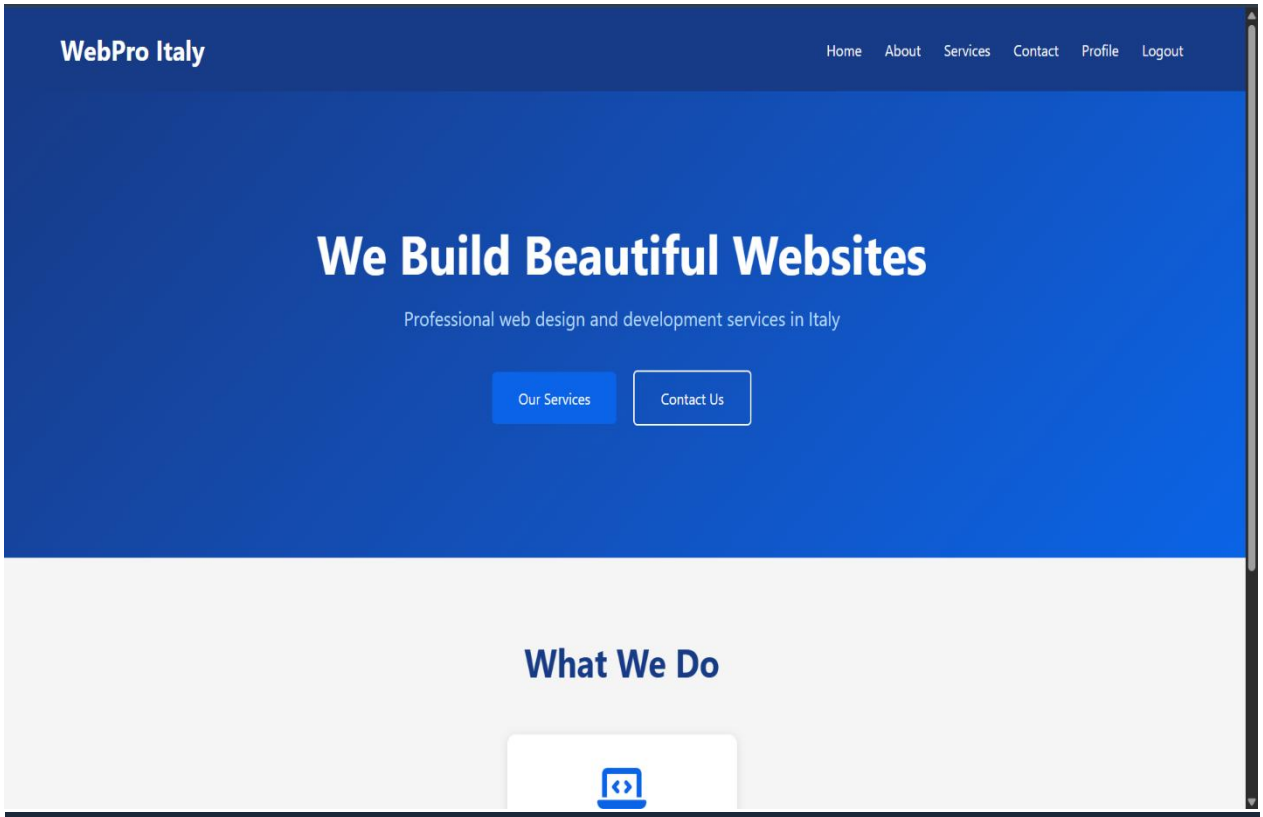
Working with sessions was also new for me. Understanding that HTTP is stateless and sessions are how we remember who is logged in between pages was an important concept.

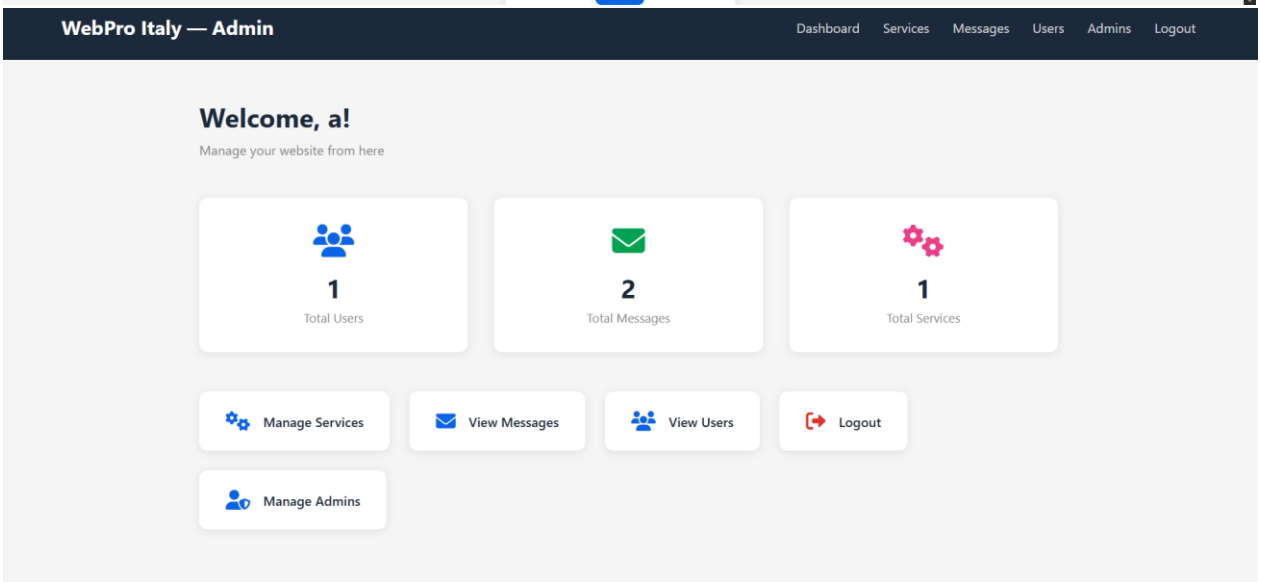
## 8. Conclusion

I built a complete web application for a web design agency with user registration, login, contact form, admin panel and API endpoints. The project works fully on a local XAMPP server and the source code is on GitHub.

If I had more time I would add more features like email notifications when a new message arrives, the ability to edit services not just add/delete them, and a better mobile responsive design.

GitHub: https://github.com/meganis10/webagency-project


WebPro Italy
Home
About
Services
Contact
Profile
Logout
We Build Beautiful Websites
Professional web design and development services in Italy
Our Services
Contact Us
What We Do


WebPro Italy — Admin
Dashboard
Services
Messages
Users
Admins
Logout
Welcome, a!
Manage your website from here
1
Total Users
2
Total Messages
1
Total Services
Manage Services
View Messages
View Users
Logout
Manage Admins

# Create Account

Your Name *

Please fill out this field.

Enter your name

Your Email *

user@gmail.com

Date of Birth *

mm/dd/yyyy

Country *

Select your country

Current City *

Enter your city

Your Password *

••••••

Confirm Password *

Confirm your password

Register Now

Already have an account? Login now

# Manage Services

Add or remove services shown on homepage

## Add New Service

Service title e.g. Web Design

Service description...

Font Awesome icon e.g. fa-laptop-code

Find icons at fontawesome.com/icons — copy the icon name e.g. fa-laptop-code

Add Service

| # | Icon | Title | Description | Action |
|---|---|---|---|---|
| 1 | | web mange | managing users... | Delete |
| 2 | | website for institutes | we are going to make your online business and increase your ... | Delete |

**WebPro Italy — Admin**

Dashboard Services Messages Users Admins Logout

# Messages

Messages sent from the contact form

| # | Name | Email | Subject | Date | Action |
|---|---|---|---|---|---|
| 1 | ali | user@gmail.com | business | 19 Apr 2026 | View Delete |